

Skills Summary

Diagnosing Causation Claims and Biased Study Designs

Use this reference while completing your worksheet or when studying.

Skill 1 — Association is not causation

Rule: a correlation says the two variables move together, nothing more. Before accepting a causal claim, look for a *common cause* (a third variable driving both) and for *reverse causation* (the effect driving the supposed cause). r^2 measures how much of the variation the linear fit accounts for — it does not upgrade an association into a cause.

Example: Two variables have $r = 0.80$. Find r^2 .

Step 1. Squaring r turns the strength of the association into the share of variation the fit accounts for — still a description, not a cause.

Step 2. $r^2 = 0.80^2 = 0.64$, so 64% of the variation is accounted for.

$\Rightarrow r^2 = \mathbf{0.64}$, and 36% is still unexplained — room for other causes.

Try it: Two variables have $r = 0.60$. Find r^2 .

check: $r^2 = 0.36$

Skill 2 — Who was actually asked

Rule: a proportion describes the group it was computed from. Ask where the sample came from: *voluntary response* (people chose to reply) over-represents strong opinions; *convenience sampling* (whoever was nearby) over-represents whoever gathers there; *undercoverage* leaves part of the population unreachable. Only a random sample of the whole population supports a claim about the whole population.

Example: Of 200 people leaving a concert, 180 said they enjoy live music. Find the sample proportion.

Step 1. The figure asked for describes the people actually asked, so the denominator is the sample size, not the town.

Step 2. $P = \frac{180}{200}$, which reduces by 20.

$\Rightarrow \frac{9}{10}$ of *those asked* — a convenience sample of concertgoers, so it says nothing about the town.

Try it: Of 150 people leaving a library, 120 said they read daily. Find the sample proportion.

check: $\frac{4}{5}$

Skill 3 — Observational study or experiment?

Rule: if the subjects chose their own group, the study is *observational* and a group difference may come from a *confounding* variable rather than the treatment. Only an *experiment* — the researcher assigns treatments at *random* — balances the unknown variables and supports a causal conclusion. Blocking first, then randomizing inside each block, handles a variable you already know matters.

Example: Five volunteers who took a training scored 61, 63, 65, 67, 69. Find the group's mean.

Step 1. Comparing groups starts with a number for each, so compute this group's mean before asking what the comparison is worth.

Step 2. $\frac{61 + 63 + 65 + 67 + 69}{5} = \frac{325}{5}$.

$\Rightarrow$ mean = **65** — but the volunteers self-selected, so the gap is not yet evidence.

Try it: The five who did not train scored 54, 56, 58, 60, 62. Find their mean.

check: 89

Skill 4 — Relative change versus the base rate

Rule: “risk cut by half” is a *relative* change; the number of people affected is the *absolute* change, $\frac{a}{n} - \frac{b}{n}$. A large relative change on a tiny base rate is a small absolute change. Always ask “half of what?” — a percentage without its base is not information.

Example: Risk falls from 6 cases per 500 to 3 per 500. Find the absolute change in risk.

Step 1. The headline would say fifty percent; the absolute change is what tells you how many people are affected, so subtract the two rates.

Step 2. $\frac{6}{500} - \frac{3}{500} = \frac{3}{500}$.

$\Rightarrow$ $\frac{3}{500}$, that is 3 fewer cases per 500 people.

Try it: Risk falls from 8 per 400 to 2 per 400. Find the absolute change.

check: $\frac{3}{200}$